Natural Misère Quotients:
Exact Octal Traces, Tame Realization, and Grundy's Game

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Abstract

Siegel's valid-transition-table theorem characterizes abstract finite misère quotients. Natural realization asks for more: one uniform heap rule must generate the table at every heap size. We express this homogeneity as an exact quotient-valued trace recurrence. A least-rank separation argument proves that the option-value set determines its value in every reduced valid table. Hence a fixed no-split code with finite quotient has an ultimately periodic pretending function and a finite exact verification procedure. Splitting introduces an unbounded convolution, but an asserted periodic split trace still has a finite certificate.

This framework yields two exact applications. For every $n \geq 2$, the code with 2^{n-1} consecutive digits equal to 3 has quotient $\mathcal{T}_n$, the tame quotient of order $2^n + 2$; the proof gives the quotient map and the outcome strategy for arbitrary heap multiplicities. For misère Grundy's game, the quotient generated by heaps through 12 is $\mathcal{T}_2$, heap 13 enlarges it to a reduced monoid of order 12, and heap 18 enlarges it to one of order 24. These are exact submonoid theorems, not bounded indistinguishability computations. The full Grundy quotient and the uniform realization of the wild $|P| = 2$ family remain open.

1 Introduction

An abstract finite impartial game can be assembled to realize a prescribed transition table. A finite octal code cannot be assembled heap by heap: its kth digit imposes the same translation-invariant move on every sufficiently large heap. The natural realization problem is to identify the extra algebraic condition imposed by that uniform tail.

The answer developed here has three layers. First, reduced transition tables are deterministic in their option-value sets. Second, the octal rule turns that determinism into an exact recurrence for the single-heap quotient word. Third, the recurrence can be solved either uniformly, as for the tame family, or prefix by prefix with all multiplicities retained, as for Grundy's game. The main results are the following.

Theorem A. *1. A reduced valid transition table assigns at most one value to each option-value set. A finite octal code realizes a finite reduced bipartite monoid exactly when its induced heap trace is total, lies in such a table, and generates the monoid.*

2. If the code has no split bits and its quotient is finite, the heap trace is ultimately periodic. Exact realization by a fixed no-split code is therefore decidable. If a split trace is already ultimately periodic, its infinite tail is likewise certified by finitely many records.

3. For every $n \geq 2$, the code $0.\underbrace{33\cdots3}_{2^{n-1}\ \text{digits}}$ has exact quotient $\mathcal{T}_n$. Thus every tame quotient in the finite $|P| = 2$ classification has a finite-code realization.

4. *In misère Grundy's game the exact quotients generated through heaps* 12, 13, *and* 18 *have orders* 6, 12, *and* 24, *respectively, with the presentations and reduced translation tables given below.*

The first two items are a characterization and a decision theorem for their stated inputs; the third is a uniform realization theorem; the fourth is an exact prefix theorem. None asserts that split traces are automatically periodic or that the full Grundy quotient is finite. Sections 2 and 3 build the trace calculus, Section 4 realizes the tame family, and Section 5 gives the Grundy extensions. Section 6 states the formal and open boundaries.

2 Transition tables and deterministic values

Let A be a set of finite impartial games closed under options and disjunctive sum. Under misère play the terminal game is an N-position. For $G, H \in A$, put

$$G \sim_A H \quad \iff \quad o^-(G+X) = o^-(H+X) \quad \text{for every } X \in A.$$

The quotient $Q(A) = A/{\sim_A}$ is a commutative monoid. It comes with the distinguished subset P of previous-player wins. The pair (Q, P) is *reduced*: if $x \neq y$, then some z has exactly one of xz, yz in P. Also $1 \notin P$, because the terminal position is N.

Plambeck and Siegel developed this quotient method and the generalized mex rule in [3]; Siegel subsequently proved the abstract classification by valid transition tables in [5]. A transition pair is $(x, E) \in Q \times \mathrm{Pow}(Q)$, with product

$$(x, E)(y, F) = (xy, yE \cup xF). \tag{1}$$

A transition table T on (Q, P) is valid when it is complete, closed under the displayed product, parity-correct,

$$(x, E) \in T \quad \Longrightarrow \quad x \in P \iff E \neq \varnothing \text{ and } E \cap P = \varnothing, \tag{2}$$

and well-founded: there is $R : Q \to \mathbb{N}$, with $R(1) = 0$, such that every x has some $(x, E) \in T$ satisfying $R(e) < R(x)$ for all $e \in E$. Siegel's theorem says that a reduced (Q, P) with $1 \notin P$ is an abstract finite misère quotient if and only if it admits such a table.

The natural problem begins after that theorem: a code must generate its table through one translation-invariant heap rule.

2.1 Option-set determinism

The first observation converts the natural problem from a nondeterministic labeling problem into a trace recurrence.

Theorem 2.1 (Option-set determinism)**.** *Let T be a valid transition table on a reduced bipartite monoid (Q, P). If*

$$(x, E) \in T \quad \textit{and} \quad (y, E) \in T,$$

then $x = y$.

Proof. Assume $x \neq y$. Among all contexts separating x and y, choose z with $R(z)$ least. Take a descending pair $(z, F) \in T$. Suppose, after interchanging x, y if necessary, that $xz \in P$ and $yz \notin P$. Closure gives

$$(xz, zE \cup xF), \qquad (yz, zE \cup yF) \in T.$$

Parity for xz says that $zE \cup xF$ is nonempty and contains no element of P. The second option set is nonempty as well: a witness coming from zE is unchanged, while a witness xf is replaced by yf. Since $yz \notin P$, parity therefore gives some P-element in $zE \cup yF$. It cannot lie in zE, which is already contained in the P-free first option set. Hence $yf \in P$ for some $f \in F$, while $xf \notin P$. Thus f also separates x, y, but $R(f) < R(z)$, a contradiction. □

Equivalently, every valid table defines a partial function

$$\mu_T(E) = x \quad \Longleftrightarrow \quad (x, E) \in T. \tag{3}$$

This strengthening of the option-image principle is purely algebraic: it does not assume that T was first obtained from games. The theorem and its least-rank proof are kernel-checked in the Lean module `Ogdoad.MisereTransition`.

We also record the standard local obstruction in a form suited to heap traces. Define the meximal set

$$M_x = \{y \in Q : \text{there is no } z \text{ with } xz, yz \in P\}.$$

Lemma 2.2 (Meximal obstruction). *Every $(x, E) \in T$ satisfies $E \subseteq M_x$.*

Proof. If $e \in E$ and $xz, ez \in P$, choose any table pair (z, F). Closure puts $(xz, zE \cup xF)$ in T, and $ez \in zE$ is a P-valued option of the P-value xz, contradicting parity. □

In any nontrivial reduced quotient, $x \notin M_x$. Thus a particularly cheap obstruction is $x \notin E$ for every transition pair (x, E).

3 Exact octal traces

3.1 The trace criterion

Write a finite octal code as three masks. Removing k counters may take a whole heap when C_k holds, may leave one nonempty heap when A_k holds, and may leave two nonempty heaps when B_k holds. These are respectively the $1, 2, 4$ bits of the kth octal digit. Let d be the last nonzero digit, put $x_0 = 1$, and seek $x_n = \Phi(H_n)$ for $n \geq 1$. The quotient image of the option set of a heap is exactly

$$\begin{aligned} E_n = \big(\{1\} \text{ if } C_n\big) &\cup \{x_{n-k} : A_k,\ k < n\} \\ &\cup \{x_i x_{n-k-i} : B_k,\ k \leq n-2,\ 1 \leq i \leq (n-k)/2\}. \end{aligned} \tag{4}$$

Equal parts are permitted in an octal split.

Theorem 3.1 (Exact trace criterion). *Fix a finite octal code Γ and a finite reduced bipartite monoid (Q, P). Then $Q(\Gamma) \cong (Q, P)$ if and only if there are a valid table T on (Q, P) and a sequence $(x_n)_{n \geq 0}$ such that*

1. *$x_0 = 1$;*
2. *$(x_n, E_n) \in T$ for every $n \geq 1$, with E_n given by the displayed octal-trace formula;*
3. *the elements x_n generate Q.*

For fixed T and Γ, the sequence is unique whenever it exists.

Proof. Necessity follows by taking the actual transition algebra and the quotient images of the heaps. Conversely define Φ multiplicatively on heap multisets. The transition pair of any position is the product of its heap pairs. Closure of T therefore puts $(\Phi(G), \Phi''G)$ in T for every position G. Induction on total counter count, using parity, gives

$$o^{-}(G) = P \iff \Phi(G) \in P.$$

Generation makes Φ onto. Equal images are indistinguishable by the displayed equivalence; distinct images are distinguished by a context in Q, and surjectivity realizes that context by a position. Thus the fibers are exactly the indistinguishability classes. Uniqueness is Theorem 2.1 applied recursively. □

This theorem identifies the general unresolved core precisely: a finite alphabet quadratic-convolution recurrence must remain defined forever. It also separates exact certificates from bounded signatures.

3.2 Finite-state and periodic traces

Suppose every digit lies in $\{0, 1, 2, 3\}$, so B_k is always false. For $n > d$, whole-heap removal is also inactive, and

$$E_n = \{x_{n-k} : A_k,\ 1 \le k \le d\}. \tag{5}$$

Theorem 3.2 (Automatic periodicity)**.** *Let Γ be a finite no-split octal code with last nonzero digit d. If $Q(\Gamma)$ is finite, then its single-heap quotient word $x_n = \Phi(H_n)$ is ultimately periodic. A repeated state occurs among at most $|Q|^d + 1$ consecutive d-windows after the exceptional prefix.*

Proof. The ordered window

$$W_n = (x_{n-d}, \dots, x_{n-1}) \in Q^d$$

determines E_n by the preceding formula. By Theorem 2.1, E_n determines x_n. Thus W_n determines W_{n+1}. The deterministic map acts on at most $|Q|^d$ states, so one state repeats and the entire subsequent orbit is periodic. □

Corollary 3.3. *For a fixed finite no-split code and a fixed finite reduced (Q, P), exact realization is decidable. The same is true when the code length is bounded in advance.*

Indeed there are finitely many transition tables on Q; validity is a finite condition, and each induced trace either fails or enters a cycle. Generation and parity can then be checked on finite data. The existential problem over codes of unbounded length is not settled by this argument.

Splitting is the exact place where finite memory fails. The new term

$$S_m = \{x_i x_j : i, j \ge 1,\ i + j = m\}$$

depends on an unbounded convolution of the prefix. Nevertheless an already periodic trace remains finitely certifiable.

Proposition 3.4 (Periodic convolution)**.** *Assume $x_{n+p} = x_n$ for all $n \ge N$, where $N, p \ge 1$. Then*

$$S_{m+p} = S_m \qquad (m \ge 2N + p).$$

Consequently the records (x_n, E_n) of a length-d octal code are periodic for all $n \ge 2N + p + d$.

Proof. For a pair $i+j=m+p$, one of i,j is at least $N+p$; subtract p from that component and use periodicity. Conversely, for $i+j=m$, one component is at least N; add p to it. Commutativity permits reordering. The one-remainder terms are easier, and whole-heap bits are inactive past d. $\square$

Thus a proposed preperiod, period, valid table, finite trace through the convolution window, and generation check form an exact infinite-rule certificate. This is a transition-table form of the periodicity argument in [4, 3].

4 Uniform realization of the tame family

For $n \geq 2$, let

$$m = 2^{n-1}, \qquad L = m+1,$$

and let Γ_n be the finite octal code

$$\Gamma_n = 0.\underbrace{33\cdots3}_{m \text{ digits}}.$$

A move subtracts any number in $\{1,\dots,m\}$ from one heap, deleting it if the remainder is zero and otherwise leaving the one remainder heap. There are no splits.

Define

$$Q_n = \{1,a\} \sqcup \{z_v : v \in \mathbb{F}_2^n\}, \qquad P_n = \{a, z_0\}.$$

Identify $v \in \mathbb{F}_2^n$ with an integer in $[0,2^n)$ and write vector addition as XOR. Multiplication is

$$a^2 = 1, \qquad z_u z_v = z_{u \,\mathsf{xor}\, v}, \qquad a z_v = z_{v \,\mathsf{xor}\, 1}. \tag{6}$$

This is Siegel's tame quotient $\mathcal{T}_n$: its kernel is $\{z_v\} \cong (\mathbb{Z}/2)^n$, and $|Q_n| = 2^n + 2$ [3, 4].

Theorem 4.1 (Tame-family realization)**.** *For every $n \geq 2$, the exact misère quotient of Γ_n is $\mathcal{T}_n$.*

Proof. For a heap H_h, let $r(h)$ be the residue of h modulo L and define

$$\Phi(H_h) = \begin{cases} 1, & r(h) = 0, \\ a, & r(h) = 1, \\ z_{r(h)}, & 2 \leq r(h) \leq m. \end{cases} \tag{7}$$

Extend Φ multiplicatively.

We first prove the outcome criterion directly. Discard residue-zero heaps. Let $t \in \mathbb{F}_2$ be the parity of the number of residue-one heaps, and let R be the multiset of residues in $\{2,\dots,m\}$. From the displayed multiplication laws,

$$\Phi(X) \in P_n \iff \begin{cases} t = 1, & R = \varnothing, \\ t \,\mathsf{xor} \bigoplus_{r \in R} r = 0, & R \neq \varnothing. \end{cases} \tag{8}$$

Induct on total counter count. A legal move changes one residue r to a different residue s. If $R = \varnothing$ and $t = 1$, changing a residue 0 or 1 either flips the parity or creates a nonzero XOR, so every option fails this criterion. If $R \neq \varnothing$ and the displayed XOR is zero, there must be at least two residues at least 2: one such residue cannot cancel either 0 or 1. A move changes the XOR by

$r \operatorname{xor} s \neq 0$, and at least one large residue remains. Hence every option of a predicted P-position is predicted N.

Conversely suppose the position is predicted N. If $R = \varnothing$ and $t = 0$, move a residue-one heap to residue zero when one exists. Otherwise the nonempty position has a positive residue-zero heap; subtracting m moves it to residue one. Either move makes $t = 1$.

Now suppose at least two residues at least 2 occur, and write

$$w = t \operatorname{xor} \bigoplus_{r \in R} r \neq 0.$$

Choose a component residue r whose binary expansion contains the highest set bit of w and put $s = r \operatorname{xor} w$. Such a component exists among the residues contributing to the XOR: it may be a residue-one component when the highest bit is the low bit, and otherwise it lies in R. Then $s < r$, so subtracting $r - s \leq m$ is legal, the new XOR is zero, and at least one large residue remains. Finally, if exactly one large residue r occurs, move it to residue $1 - t < r$. No large residue remains and the new residue-one parity is odd. This proves the displayed criterion is the actual misère outcome for every position.

The map is onto. The heap residues provide a and $z_2, z_4, \ldots, z_m$; also $z_2^2 = z_0$ and $az_0 = z_1$. These are a binary basis for every z_v. Finally (Q_n, P_n) is reduced. Distinct z_u, z_v are separated by z_u. The elements $1, a$ are separated by 1; the context a separates 1 from z_v unless $v = 1$, when z_1 does; and the context 1 separates a from z_v unless $v = 0$, when z_0 does. Outcome correctness, surjectivity, and reduction identify the fibers of Φ with exact indistinguishability classes. $\square$

For $n = 2$, the construction is 0.33 and gives the six-element quotient $\mathcal{T}_2$. The theorem realizes the entire tame arm in Siegel's classification of finite quotients with $|P| = 2$; the other arm consists of the tame extensions of the eight-element wild quotient $\mathcal{R}_8$ [5]. The base $\mathcal{R}_8$ itself is realized by 0.75 [3], but a uniform finite-code realization of all its tame extensions is not supplied here.

5 Exact initial quotients of misère Grundy's game

In Grundy's game a heap may be split into two unequal positive heaps:

$$\operatorname{opts}(H_n) = \{H_i + H_{n-i} : 1 \leq i < n - i\}.$$

The rule preserves total counters but decreases the well-founded quantity $\sum_h \max(h - 2, 0)$. Its misère form is a longstanding difficult example; the early analysis goes back to Grundy and Smith [2] and is discussed in [1].

5.1 The tame prefix through heap 12

We first give an exact transition certificate through heap 12. Write

$$\mathcal{T}_2 = \langle a, b \mid a^2 = 1,\ b^3 = b \rangle, \qquad P = \{a, b^2\}.$$

The heap values are

$$x_1 = x_2 = 1, \qquad x_n = \begin{cases} a, & n \equiv 0 \pmod 3, \\ 1, & n \equiv 1 \pmod 3, \\ b, & n \equiv 2 \pmod 3 \end{cases} \quad (3 \leq n \leq 12).$$

Their distinct transition records are

$$(1,\varnothing),\ (a,\{1\}),\ (1,\{a\}),\ (b,\{1,a\}),$$
$$(a,\{1,b\}),\ (1,\{a,b\}),\ (b,\{1,a,ab\}).$$

Closing these records under the transition product gives the following 24 pairs; each row lists the option sets paired with its left-hand value.

1	$\varnothing$; $\{a\}$; $\{a,b\}$; $\{a,ab\}$; $\{a,b,ab\}$
b	$\{1,a\}$; $\{b^2,ab^2\}$; $\{1,a,ab\}$; $\{b^2,ab,ab^2\}$; $\{1,b^2,a,ab\}$; $\{1,a,ab,ab^2\}$; $\{1,b^2,a,ab,ab^2\}$
b^2	$\{b,ab\}$; $\{b,ab,ab^2\}$
a	$\{1\}$; $\{1,b\}$; $\{1,ab\}$; $\{1,b,ab\}$
ab	$\{1,b,a\}$; $\{b,b^2,ab^2\}$; $\{1,b,b^2,a\}$; $\{1,b,a,ab^2\}$; $\{1,b,b^2,a,ab^2\}$
ab^2	$\{b,b^2,ab\}$

The table is closed and every pair obeys parity; a,b occur as heap values and $\mathcal{T}_2$ is reduced. The trace criterion therefore proves:

Proposition 5.1. *The exact misère quotient generated by $H_1,\ldots,H_{12}$ is $\mathcal{T}_2$.*

5.2 The first extension at heap 13

Before extending this quotient, we isolate a substitution point that is easy to miss. Write $G \equiv_- H$ when G and H have the same misère outcome after addition of every finite impartial game, not merely every game in the present heap universe.

Lemma 5.2 (Global substitution)**.** *Let U be the additive closure of games G_i, and suppose that each G_i is globally equivalent to a representative R_i lying in an additively closed subuniverse $V \subseteq U$. Then inclusion induces an isomorphism $Q(V) \cong Q(U)$.*

Proof. Global equivalence is an additive congruence, so every $X \in U$ is globally equivalent to the sum $r(X) \in V$ obtained by replacing its components. If $A, B \in V$ agree against all V-contexts, then for every $X \in U$,

$$o^-(A+X) = o^-(A+r(X)) = o^-(B+r(X)) = o^-(B+X).$$

The converse follows because $V \subseteq U$, and every U-class has a representative in V. □

The Grundy–Smith canonical reduction rules are global reductions. Applied to the explicit option lists on heaps through 12, they give

$$\begin{array}{r|l} 0 & H_1, H_2, H_4, H_7, H_{10} \\ a = *1 & H_3, H_6, H_9, H_{12} \\ b = *2 & H_5, H_8, H_{11}. \end{array}$$

These are the reductions displayed in [1, pp. 419–421]. Consequently adjoining H_{13} cannot secretly refine one of these earlier identifications: they remain valid against the new context H_{13} and all of its sums.

Heap 13 has option-value set $\{a,b,b^2\}$. The alleged continuation $x_{13} = 1$ is impossible: multiplying its transition by two copies of the H_5 transition produces

$$(b^2, \{b, b^2, ab, ab^2\}),$$

which violates parity because both the value and one option lie in P. Concretely $H_4 + 2H_5$ is P, while $H_{13} + 2H_5$ has the P-option $H_8 + 3H_5$ and is N.

The extension can still be solved symbolically. Let c denote the new heap value, with transition $c \to \{a, b, b^2\}$. For raw monomials $a^e b^j c^k$, direct induction on (k, j, e) gives the following complete outcome law:

$$a^e b^j c^k \in P \iff \begin{cases} k \text{ even},\ e \text{ even},\ j > 0 \text{ even}; & \text{or} \\ k \text{ even},\ e \text{ odd},\ j = 0; & \text{or} \\ k \text{ odd},\ e \text{ odd},\ j = 0 \text{ or } j \geq 3 \text{ odd}. \end{cases} \tag{9}$$

For completeness, the induction uses only

$$a \to 1, \qquad b \to \{1, a\}, \qquad c \to \{a, b, b^2\}.$$

Here is the complete case induction. Put $\epsilon = e \bmod 2$. For a listed P-state with k even, either $(\epsilon, j) = (0, \text{positive even})$ or $(1, 0)$; the a, b, c moves land respectively in the complementary parity classes, and the three c replacements land at $(1, j), (0, j+1), (0, j+2)$ with k odd. None is listed. For a listed state with k odd, $\epsilon = 1$ and $j = 0$ or odd at least 3; the a move sets $\epsilon = 0$, the b moves make j even, and the c moves land at $(0, j), (1, j+1), (1, j+2)$ with k even. Again none is listed.

Conversely, every unlisted nonterminal state has the following listed option. When k is even:

ϵ	j	move to a P-state
0	0	$c \to a$ (unless $k = 0$, when the position is terminal)
0	1	$b \to a$
0	odd ≥ 3	$b \to 1$
1	positive even	$a \to 1$
1	1	$b \to 1$
1	odd ≥ 3	$b \to a$

When k is odd, the witnesses are

ϵ	j	move	ϵ	j	move
0	0	$c \to a$	0	positive even	$c \to b^2$
0	odd	$c \to b$	1	1	$b \to 1$
1	2	$c \to a$	1	even ≥ 4	$b \to 1$

The omitted odd-k cases are exactly the listed P cases. This proves the misère recursion for all multiplicities; no exponent cutoff is used.

The outcome law also proves the presentation rather than merely suggesting it. It first factors raw monomials through the sixteen forms $a^\epsilon b^j c^\eta$, with $\epsilon, \eta \in \{0, 1\}$ and $j \in \{0, 1, 2, 3\}$, using $a^2 = c^2 = 1$ and $b^4 = b^2$. It is invariant under $b^2 \sim ab^3c$. The congruence generated by this relation pairs the eight forms in the b^2-ideal into four two-element orbits and leaves the eight forms with $j = 0, 1$ fixed. Thus it has exactly twelve candidate classes. The resulting quotient has presentation

$$Q_{13} = \langle a, b, c \mid a^2 = c^2 = 1,\ b^4 = b^2,\ b^2 = ab^3c \rangle, \qquad P_{13} = \{a, ac, b^2\}. \tag{10}$$

It has the twelve normal forms

$$1, c, b, bc, b^2, b^2c, b^3, b^3c, a, ac, ab, abc.$$

Reduction is visible from their distinct P-translation rows:

1	$\{b^2, a, ac\}$	c	$\{b^2c, a, ac\}$
b	$\{b, b^3\}$	bc	$\{bc, b^3c\}$
b^2	$\{1, b^2, abc\}$	b^2c	$\{c, b^2c, ab\}$
b^3	$\{b, b^3, ac\}$	b^3c	$\{bc, b^3c, a\}$
a	$\{1, c, b^3c\}$	ac	$\{1, c, b^3\}$
ab	$\{b^2c, ab\}$	abc	$\{b^2, abc\}$

Here the row at x is $\{z : xz \in P_{13}\}$. All rows differ, so the bipartite monoid is reduced. Thus the all-multiplicity outcome law proves:

Proposition 5.3. *The exact misère quotient generated by $H_1, \ldots, H_{13}$ is the twelve-element reduced bipartite monoid in the preceding presentation.*

5.3 The second extension at heap 18

The next heaps have values

$$x_{14} = b, \quad x_{15} = a, \quad x_{16} = c, \quad x_{17} = b.$$

Here these assertions are again global, not guesses made inside Q_{13}. With 0 denoting the terminal game, the split rule gives

$$\mathrm{opts}(H_{14}) = \{c, ab, a, 0\}, \qquad \mathrm{opts}(H_{15}) = \{c, b, 0\},$$
$$\mathrm{opts}(H_{16}) = \{ac, b^2, b, a\}, \qquad \mathrm{opts}(H_{17}) = \{c, ab, a, 0\}.$$

The reversible-option reductions of Grundy and Smith give respectively $H_{14} \equiv_- b$, $H_{15} \equiv_- a$, $H_{16} \equiv_- c$, and $H_{17} \equiv_- b$; these are also the four reductions printed in [1, p. 421]. Heap 18 has exact option set $\{0, b, c, bc\}$ and creates a generator $d = H_{18}$. By Lemma 5.2, every position generated by $H_1, \ldots, H_{18}$ is globally equivalent to a count state $a^e b^j c^k d^r$, and every such count state occurs in that universe.

The all-multiplicity recursion extends once more:

$$\begin{aligned} a^e b^j c^k d^r \in P \quad \iff \quad & \big(r \text{ even and } a^e b^j c^k \text{ satisfies the} \\ & \qquad \text{preceding three-generator law}\big) \\ & \text{or } \big(r \text{ odd, } e \text{ even, } j = 0\big). \end{aligned} \tag{11}$$

To check the recursion, use $d \to \{1, b, c, bc\}$. If r is even and the three-generator state is P, the preceding proof handles the a, b, c moves, while every d move has odd r and either odd e or positive j. If the three-generator state is N, use its displayed winning move. The only case with no a, b, c component is terminal when $r = 0$; for positive even r, the move $d \to 1$ reaches the odd-r clause.

If r is odd, a state in the displayed P clause has even e and $j = 0$. An a move makes e odd, a c move makes either e odd or j positive, and a d move lands, at even r, in one of

$$(\epsilon, j, k) = (0, 0, k), (0, 1, k), (0, 0, k+1), (0, 1, k+1),$$

all of which fail the three-generator criterion. Conversely an odd-r N-state has a P move in the following exhaustive cases:

ϵ	j	move	ϵ	j	move
1	0	$a \to 1$	0	1	$b \to 1$
0	even ≥ 2	$d \to 1$ if k even; $d \to c$ if k odd	0	odd ≥ 3	$d \to b$ if k even; $d \to bc$ if k odd
1	1	$b \to a$	1	even ≥ 2	$d \to bc$ if k even; $d \to b$ if k odd
1	odd ≥ 3	$d \to c$ if k even; $d \to 1$ if k odd			

Each target is one of the three even-r cases already proved. This establishes the outcome formula for all $(e, j, k, r) \in \mathbb{N}^4$.

As before, the formula itself identifies the finite presentation. Before the last relation there are 32 forms $a^\epsilon b^j c^\eta d^\rho$, with $\epsilon, \eta, \rho \in \{0, 1\}$ and $0 \le j \le 3$, subject to $a^2 = c^2 = d^2 = 1$ and $b^4 = b^2$. The relation $b^2 = ab^3c$ acts freely in the sixteen-element b^2-ideal, producing eight pairs, and leaves the sixteen forms with $j = 0, 1$ fixed. Hence exactly 24 candidate forms remain, and the outcome law descends to them. The quotient is

$$Q_{18} = Q_{13} \times \langle d \mid d^2 = 1 \rangle, \qquad P_{18} = \{a, ac, b^2, d, cd\}. \tag{12}$$

Its 24 normal forms are the twelve above and their products by d. For an explicit reduction certificate, let R be the ordered set of twelve base forms, put $P_0 = \{a, ac, b^2\}$ and $P_1 = \{1, c\}$, and define

$$A_q = \{z \in R : qz \in P_0\}, \qquad B_q = \{z \in R : qz \in P_1\}.$$

The first coordinate A_q is the Q_{13} row already displayed. The second coordinates are

q	B_q	q	B_q	q	B_q
1	$\{1, c\}$	c	$\{1, c\}$	b	$\varnothing$
bc	$\varnothing$	b^2	$\varnothing$	b^2c	$\varnothing$
b^3	$\varnothing$	b^3c	$\varnothing$	a	$\{a, ac\}$
ac	$\{a, ac\}$	ab	$\varnothing$	abc	$\varnothing$

The translation row of q in Q_{18} is (A_q, B_q), while that of qd is (B_q, A_q). The twelve displayed pairs and their twelve swaps are all distinct, so every two normal forms have a separating context. Hence:

Theorem 5.4 (Exact Grundy prefix)**.** *The exact misère quotient generated by $H_1, \ldots, H_{18}$ is the 24-element reduced bipartite monoid in the preceding presentation.*

The statements concern entire submonoids generated by the indicated heaps; no multiplicity bound is present. They do not decide whether the full Grundy quotient is finite. They do show that the apparent period-three tame pattern fails for an exact, explicit reason at heap 13, and they give the first two symbolic extension layers for a future infinite-family or stabilization argument.

Only the individual-heap reductions displayed above are global. The relations $c^2 = d^2 = 1$, $b^4 = b^2$, and $b^2 = ab^3c$ are conclusions inside the localized quotient Q_{18}; they are not asserted as identities in the global semigroup of all misère impartial games. (The relation $a^2 = 1$ is global.)

6 Formal verification and open boundary

The results divide the original problem into sharper pieces.

1. Every tame finite quotient $\mathcal{T}_n$ has a uniform finite-code realization. For the $|P| = 2$ classification, the natural realization question remains for the family obtained by tame extension of $\mathcal{R}_8$.

2. For no-split codes, finite quotient implies automatic periodicity and exact verification is finite-state. With splitting, the unresolved object is the totality and synthesis of the finite-alphabet convolution trace displayed in Section 3.

3. For Grundy's game, Q_{13} and Q_{18} are exact. A solution of the full problem must either continue these presentations uniformly and prove stabilization, or produce an infinite family of distinct translation rows with explicit separating contexts.

The import-only entry point `Ogdoad.Papers.MisereNaturalRealization` exposes the Lean development, which currently checks the load-bearing transition determinism theorem, the meximal obstruction, and the periodic split- convolution lemma. The tame-family strategy and the Grundy prefix presentations are proved in this note at paper level; their finite monoid models and exponent recursions are the next natural formalization targets.

7 Conclusion

Natural realization is not an additional finite-table axiom. It is a totality problem for a quotient-valued heap trace. Transition-table determinism makes that trace canonical: no-split codes become finite-state, while split codes expose exactly the convolution that prevents the same automatic periodicity argument.

Within this framework the tame family is complete. One explicit sequence of finite codes realizes every $\mathcal{T}_n$, with an outcome proof valid for all positions. Grundy's game exhibits the complementary phenomenon: its tame prefix breaks at heap 13 and extends again at heap 18, but each finite stage remains exactly computable as a reduced all-multiplicity quotient. A complete solution must now control that extension process uniformly—either by eventual stabilization or by explicit separating contexts for infinitely many classes.

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